Multimedia Communications
Sample Exam Answers and Study Notes

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# 1 Multimedia Representation and Perception

**Domanda 1**

Explain the Contrast Sensitivity Function (CSF): what it is, in which units spatial frequency is measured, where it peaks, and the direct implication for quantization in compression.

- **CSF**: HVS sensitivity to luminance contrast as spatial frequency changes.
- **Unit**: **cycles per degree** of visual angle.
- **Shape**: band-pass $\rightarrow$ low sensitivity at very low and very high frequencies; peak at intermediate frequencies.
- **Compression consequence** $\rightarrow$ frequency-dependent quantization:
  - **visually important frequencies** $\rightarrow$ smaller steps;
  - **less visible frequencies** $\rightarrow$ coarser steps.

**Domanda 2**

Compare cones and rods (number, function, lighting conditions) and explain why the RGB-to-Y conversion weights the green component the most.

- **Rods**: high light sensitivity, low-light vision, no color.
- **Cones**: color vision, brighter conditions, short/medium/long wavelength classes.
- **Luminance example**:

$$Y \approx 0.299R + 0.587G + 0.114B$$

- **Green coefficient largest** $\rightarrow$ HVS most sensitive to green/yellow luminance detail.

**Domanda 3**

Explain the J:a:b chroma subsampling notation. Compute the data-reduction factor of 4:2:0 vs full RGB and justify why it is perceptually acceptable.

- $J : a : b$ $\rightarrow$ chroma samples kept relative to $J$ luma samples over two rows.
- **4:2:0** $\rightarrow$ full-resolution $Y$, half horizontal and half vertical $Cb, Cr$.
- Over $2 \times 2$ pixels:
  - **RGB**: $4 \times 3 = 12$ component samples;
  - **YCbCr 4:2:0**: $4Y + 1Cb + 1Cr = 6$ component samples.
- **Reduction** $\rightarrow$ about $1/2$ of full RGB component samples.

- **Perceptual reason** → eye is more sensitive to luminance detail than chrominance detail.

## Domanda 4

Draw the Basic Tools for Compression scheme (Transform -> Prediction -> Quantization -> Entropy Coding) and indicate which stage is the only lossy one and why.

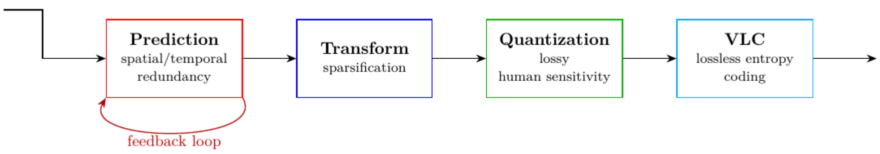


Figure 1: Pasted image 20260624205523.png

- **Prediction / transform** → decorrelate signal or compact energy.
- **Entropy coding** → lossless; shorter codes for more probable symbols.
- **Quantization** → only lossy stage: many input values map to same reconstruction value → exact original cannot be recovered.

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## 2 Compression, Prediction and Lossless Coding

**Domanda 5**

Principles of image compression. Discuss the criteria for evaluating a compression algorithm: rate, quality, [Bonus: robustness, delay, complexity].

- **Compression exploits**:
  - **statistical redundancy** → correlated samples/blocks/frames;
  - **spatial / temporal redundancy** → repeated structures;
  - **psychovisual redundancy** → imperceptible distortion.
- **Types**:
  - **lossless** → exact reconstruction, lower compression;
  - **lossy** → approximate but perceptually close reconstruction, higher compression.
- **Rate**:

$$R_{\text{image}} = \frac{B_{\text{out}}}{NM} \quad \text{[bpp]}$$

$$R_{\text{stream}} = \frac{B_{\text{out}}}{T} \quad \text{[bit/s]}$$

- **Compression ratio**:

$$\text{CR} = \frac{B_{\text{in}}}{B_{\text{out}}}$$

- **Distortion / quality**:

$$D(f, \hat{f}) = \frac{1}{NM}\|f - \hat{f}\|^2$$

$$\text{PSNR}(f, \hat{f}) = 10 \log_{10}\left(\frac{V^2}{D(f, \hat{f})}\right)$$

- **Perceptual weighting**:

$$D_W(f, \hat{f}) = \frac{1}{NM}\|h \star (f - \hat{f})\|^2$$

$$\text{WPSNR}(f, \hat{f}) = 10 \log_{10}\left(\frac{V^2}{D_W(f, \hat{f})}\right)$$

- **Tradeoffs** → rate, quality, robustness, delay, complexity; also SSIM, LPIPS.

## Domanda 6

A zero-mean Gaussian random signal has an autocorrelation function:

$$r_X(n-m) = E[X(n)X(m)] = \sigma^2 \rho^{|n-m|}$$

- Consider the predictor $V(n) = X(n-1)$. For which values is the prediction gain positive?
- Optimal linear predictor is $V(n) = -\sum_{i=1}^{P} a_i x_{n-i}$, with $a = -R_X^{-1} r$, $R_X(i,j) = r_X(i-j)$, $r = [r_X(1) \cdots r_X(P)]^T$. Find optimal linear predictor of order $P = 1$ and compute prediction gain, compare then with previous case.
- Compute the optimal predictor of order 2. Compare with the previous cases.

**Given**:

$$r_X(n-m) = \mathbb{E}[X(n)X(m)] = \sigma^2 \rho^{|n-m|}$$

- Predictor $V(n) = X(n-1)$:

$$\sigma_y^2 = \mathbb{E}[(X(n) - X(n-1))^2] = 2\sigma^2(1-\rho)$$

$$G_P = 10 \log_{10}\left(\frac{1}{2(1-\rho)}\right)$$

- **Positive gain if and only if**:

$$\rho > \frac{1}{2}$$

- **Optimal linear predictor**:

$$V(n) = -\sum_{i=1}^{P} a_i x(n-i), \quad \vec{a}^{\,opt} = -R_X^{-1}\vec{r}$$

- For $P = 1$:

$$a_1^{opt} = -\rho \quad \Rightarrow \quad V(n) = \rho X(n-1)$$

$$\sigma_y^2 = \sigma^2(1-\rho^2), \quad G_P = 10 \log_{10}\left(\frac{1}{1-\rho^2}\right)$$

- For $P = 2$:

$$R_X = \sigma^2 \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}, \quad \vec{r} = \sigma^2 \begin{bmatrix} \rho \\ \rho^2 \end{bmatrix}$$

$$\vec{a}^{\,opt} = \begin{bmatrix} -\rho \\ 0 \end{bmatrix}$$

- **Result** $\rightarrow$ second tap gives no extra gain for AR(1) source.

**Domanda 7**

Draw and comment on the schemes of predictive quantization (encoder and decoder).

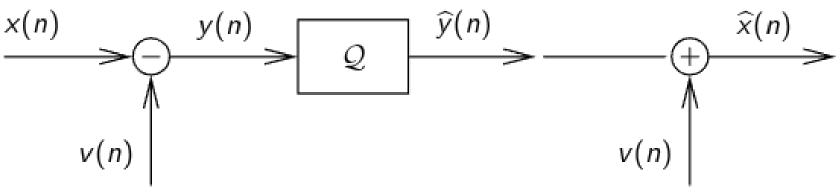


Figure 2: Predictive quantization - open loop.png

- **Open-loop predictive quantization**:
  - subtract predictor $v(n)$ from $x(n)$ → residual $y(n)$;
  - **quantize residual** → $\hat{y}(n)$;
  - add $v(n)$ back → reconstruction $\hat{x}(n)$.
- **Problem** → encoder predicts from original samples, decoder from reconstructed samples → **drift**.

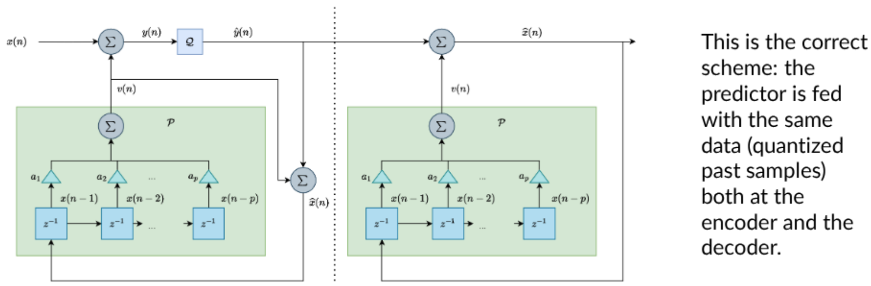


Figure 3: Predictive quantization - correct closed loop.png

- **Closed-loop encoder** → encoder contains same reconstruction loop as decoder.
- **Predictor input matches both sides**:

$$y(n) = x(n) - v(n), \quad \hat{x}(n) = \hat{y}(n) + v(n)$$

- Useful if and only if residual variance is smaller than original variance.

**Domanda 8**

Discuss the principles of lossless coding.

- **Lossless coding** → maps symbols to bitstrings; reconstructs original sequence exactly.

- **Alphabet/code**:

$$\mathcal{X} = \{x_1, \ldots, x_M\}, \quad \mathcal{C} : \mathcal{X} \to \{0,1\}^*$$

- **Fixed-length coding**:

$$R = \lceil \log_2 M \rceil$$

- **VLC** $\to$ probable symbols get shorter codewords:

$$\bar{L} = \sum_i p_i l_i$$

- **Prefix code** $\to$ no codeword is prefix of another $\to$ instantaneous decoding.
- **McMillan theorem** $\to$ best uniquely decodable code has same performance as best prefix code.
- **Kraft inequality**:

$$\sum_i 2^{-l_i} \leq 1 \iff \text{prefix code exists with lengths } \{l_i\}$$

- **Entropy lower bound**:

$$H(X) = -\sum_i p_i \log_2 p_i$$

- **Shannon**:

$$H(X) \leq \bar{L} < H(X) + 1$$

**Domanda 9**

Consider a source that emits the symbols A, B, C, D, E, and F. The probabilities of these symbols are given in the following table:

$$p_A = 0.3,\ p_B = 0.1,\ p_C = 0.05,\ p_D = 0.18,\ p_E = 0.15,\ p_F = 0.22$$

- Describe the Huffman Algorithm.
- Compute Huffman code for this distribution.
- Compare the average length of the Huffman code with the distribution entropy.

- **Huffman algorithm** $\to$ repeatedly merge two least probable symbols.
- **One optimal code**:
  - A, $p = 0.30 \to$ '10', length 2
  - B, $p = 0.10 \to$ '1111', length 4
  - C, $p = 0.05 \to$ '1110', length 4
  - D, $p = 0.18 \to$ '00', length 2
  - E, $p = 0.15 \to$ '110', length 3

   - F, $p = 0.22 \to$ '01', length 2

$$\bar{L} = 0.30 \cdot 2 + 0.10 \cdot 4 + 0.05 \cdot 4 + 0.18 \cdot 2 + 0.15 \cdot 3 + 0.22 \cdot 2 = 2.45$$

$$H(X) = -\sum_i p_i \log_2 p_i \approx 2.406 \text{ bit/symbol}$$

- **Overhead** $\to \bar{L} - H(X) \approx 0.044$ bit/symbol.

**Domanda 10**

Why arithmetic encoder is preferred over Huffman for high-performance lossless coding?

- **Huffman** $\to$ integer codeword lengths $\to$ penalty for non-dyadic probabilities.
- Block Huffman reduces penalty, but alphabet grows as $M^K$.
- **Arithmetic coding** $\to$ encodes whole sequence as interval in $[0, 1)$.
- **Rate**:

$$\mathcal{L} < H(X) + \frac{2}{n} \xrightarrow[n\to\infty]{} H(X)$$

- Also suited to adaptive/context probability models.

**Domanda 11**

(LLCod) Explain the dif and only iference between Fixed-Length Coding (FLC) and Variable-Length Coding (VLC), and describe why VLC is theoretically superior for non-equiprobable sources.

- **FLC** $\to$ same codeword length for every symbol; simple, instant parsing, inefficient if probabilities non-uniform.
- **VLC** $\to$ dif and only iferent lengths; probable symbols shorter.

$$\bar{L} = \sum_i p_i l_i$$

- **Superiority condition** $\to$ non-equiprobable source.

**Domanda 12**

(LLCod) Discuss the importance of the prefix condition in Variable-Length Coding and how it relates to the concept of instantaneous decodability.

- **Prefix condition** $\to$ no codeword is prefix of another.

- **Consequence** → decoder identifies symbol as soon as codeword ends; no look-ahead.
- **Kraft**:

$$\sum_i 2^{-l_i} \leq 1$$

- **Equality** → complete prefix tree.

**Domanda 13**

(LLCod) What are the two distinct mechanisms by which "block coding" improves the efficiency of lossless compression?

1. **Sources with memory**:

$$H(X^K) \leq \sum_{i=1}^{K} H(X_i)$$

→ dependencies exploited.

1. **Memoryless non-dyadic sources**:

$$\frac{H(X^K)}{K} \leq \frac{L^*}{K} < \frac{H(X^K)}{K} + \frac{1}{K}$$

→ integer-length overhead spread over $K$ symbols; overhead/symbol → 0.

**Domanda 14**

(LLCod) Provide a synthetic comparison between the main lossless coding techniques (Exp-Golomb, Huffman, Arithmetic, Dictionary, Neural) in terms of complexity and Latency, and provide a typical use case for each of them.

- **Exp-Golomb**: good for small integers; very low complexity/latency → syntax elements, motion-vector residuals.
- **Huffman**: optimal among symbol prefix codes; low/medium complexity, very low latency → JPEG, DEFLATE-style coding.
- **Arithmetic**: near entropy; high complexity, medium latency → CABAC, context-adaptive coding.
- **Dictionary (LZ/LZW)**: universal for repeated patterns; medium complexity, low/medium latency → text, GIF, ZIP-like systems.
- **Neural lossless**: potentially very high efficiency; very high complexity/latency → research, high-resolution image models.

## Domanda 15

(LLCod) Describe the principle of the Huffman coding. For the following probability distribution, compute the optimal lossless code, and compare its average length to the source's entropy:

$$p_A = 0.35,\ p_B = 0.1,\ p_C = 0.07,\ p_D = 0.08,\ p_E = 0.12,\ p_F = 0.28$$

- **One optimal Huffman code**:
  - A, $p = 0.35 \to$ '00', length 2
  - B, $p = 0.10 \to$ '100', length 3
  - C, $p = 0.07 \to$ '101', length 3
  - D, $p = 0.08 \to$ '110', length 3
  - E, $p = 0.12 \to$ '111', length 3
  - F, $p = 0.28 \to$ '01', length 2
- **Relevant lengths**:

$$l_A = l_F = 2, \quad l_B = l_C = l_D = l_E = 3$$

$$\bar{L} = 0.35 \cdot 2 + 0.28 \cdot 2 + (0.10 + 0.07 + 0.08 + 0.12) \cdot 3 = 2.37$$

$$H(X) \approx 2.304 \text{ bit/symbol}$$

- **Overhead** $\to \bar{L} - H(X) \approx 0.066$ bit/symbol.

## Domanda 16

(LLCod) Which of the following statements best describes the behavior of the Shannon entropy $H(X)$ for a binary random variable with probability $p$?

For $X \sim \text{Bernoulli}(p)$:

$$H(X) = -p\log_2 p - (1-p)\log_2(1-p)$$

- $H(X) = 0$ for $p = 0$ or $p = 1$.
- Maximum at $p = \frac{1}{2}$:

$$H\left(\frac{1}{2}\right) = 1 \text{ bit}$$

## Domanda 17

(LLCod) Why is Lempel-Ziv (e.g., LZW) considered a "universal" coding algorithm?

- No prior source probability distribution needed.
- Encoder and decoder build same dictionary adaptively from observed sequence.
- **Stationary ergodic sources** → asymptotically optimal.

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# 3 Transform Coding and JPEG

**Domanda 18**

Why the geometric mean of the variances of a random vector is key information to evaluate the rate-distortion performance of a quantizer?

- High-resolution transform coding with optimal bit allocation:

$$D^\star = c_{GM}\sigma_{GM}^2 2^{-2\bar{R}}$$

- Orthogonal transforms preserve total energy → preserve arithmetic mean:

$$\sigma_{AM,Y}^2 = \sigma_{AM,X}^2$$

- Transform can change variance distribution.
- **Good transform** → few high-energy coefficients, many low-energy coefficients → lower $\sigma_{GM}^2$.

$$\sigma_{GM}^2 \leq \sigma_{AM}^2$$

- **Lower geometric mean** → lower distortion at same rate.

$$G_T = \frac{D_{\text{PCM}}}{D_Y} = \frac{\sigma_{AM,Y}^2}{\sigma_{GM,Y}^2}$$

$$G_{T,dB} = 10\log_{10} G_T$$

**Domanda 19**

Write the resource allocation problem for transform coding. Derive the Huang-Schulteiss formula.

- **High-resolution distortion**:

$$D = \frac{1}{M}\sum_{k=0}^{M-1} c_k\sigma_k^2 2^{-2R_k}$$

- **Rate constraint**:

$$\sum_{k=0}^{M-1} R_k \leq R_{\text{tot}}$$

- **Lagrangian**:

$$J(\vec{R},\lambda) = \frac{1}{M}\sum_{k=0}^{M-1} c_k\sigma_k^2 2^{-2R_k} + \lambda\left(\sum_{k=0}^{M-1} R_k - R_{\text{tot}}\right)$$

- Set $\frac{\partial J}{\partial R_k} = 0$ → Huang-Schultheiss:

$$R_k^\star = \frac{R_{\text{tot}}}{M} + \frac{1}{2}\log_2\left(\frac{c_k\sigma_k^2}{c_{GM}\sigma_{GM}^2}\right)$$

$$c_{GM} = \sqrt[M]{\prod_{k=0}^{M-1} c_k}, \quad \sigma_{GM}^2 = \sqrt[M]{\prod_{k=0}^{M-1} \sigma_k^2}$$

- **Above geometric mean** → more bits.
- **Below geometric mean** → fewer bits.

**Domanda 20**

Explain why the KLT is the optimal linear transform for decorrelation and why the DCT is used in practice instead.

- **KLT** → eigenvectors of signal covariance matrix.
- **In KLT basis** → coefficients decorrelated; optimal energy compaction among linear orthogonal transforms for given statistics.

$$T_{KLT} = [u_1 \ u_2 \ \cdots \ u_M]^T$$

- **Problem** → KLT depends on actual image/block statistics; encoder must estimate covariance, compute transform, signal it.
- **DCT** used because fixed, separable, fast, no side information, good KLT approximation for locally correlated image blocks.

**Domanda 21**

Describe the principles of the JPEG standard.

- **JPEG** → lossy still-image codec based on block DCT, quantization, entropy coding.
- Standard mainly defines decodable bitstream and decoder behavior; encoder choices flexible.

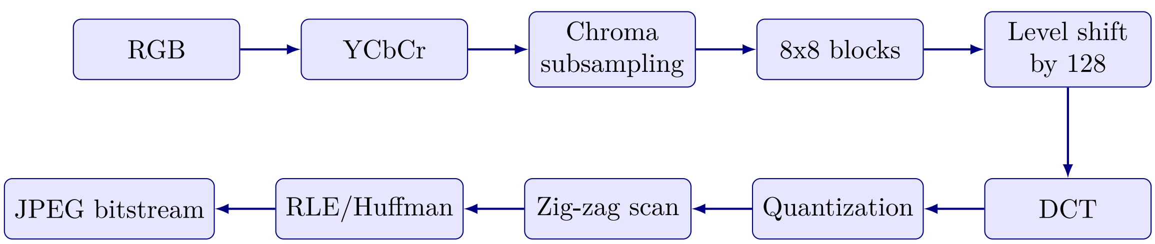


Figure 4: JPEG Coding Pipeline

- **Quantization**:

$$\tilde{C}_{ij} = \text{round}\left(\frac{C_{ij}}{q_{ij}}\right)$$

- **Quantization table** → smaller steps for important low frequencies, larger steps for high frequencies.
- **Quality factor**:

$$S_F = \begin{cases} \frac{5000}{Q} & 1 \leq Q \leq 50 \\ 200 - 2Q & 50 < Q \leq 99 \\ 1 & Q = 100 \end{cases}$$

$$q \leftarrow \frac{S_F}{100} q^{\star}$$

- **Metadata**:
  - **JFIF** → density, resolution, thumbnails;
  - **EXIF** → date/time, GPS, camera settings.

**Domanda 22**

(TC&JPEG) Draw the scheme and describe the functional blocks of a JPEG encoder.

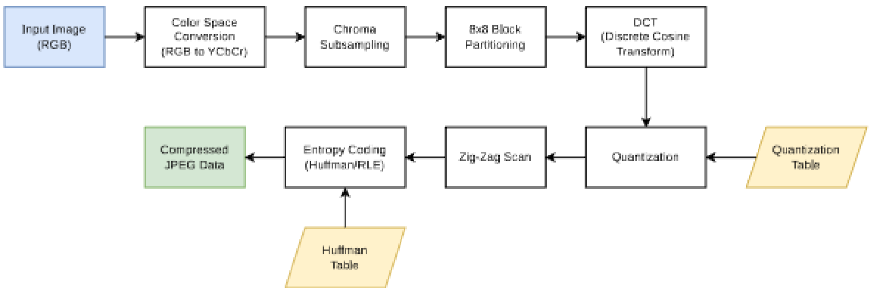


Figure 5: JPEG encoder.png

- **DCT** → energy compaction.
- **Quantization** → lossy coefficient reduction.
- **Zig-zag scan** → groups zeros.
- **Entropy coding** → lossless compression.

**Domanda 23**

(TC&JPEG) Describe the problem of “frequency leakage” in the Discrete Fourier Transform (DFT) when applied to signal compression and how the Discrete Cosine Transform (DCT) addresses it.

- DFT assumes finite signal is periodic.
- **Boundary mismatch** $\rightarrow$ artificial discontinuities $\rightarrow$ energy spread to high frequencies = **frequency leakage**.
- **DCT uses symmetric extension** $\rightarrow$ smoother periodic continuation $\rightarrow$ better energy compaction for images.

### Domanda 24

(TC&JPEG) Explain the entropy coding process for AC coefficients in the JPEG standard and the significance of the "End of Block" (EOB) symbol.

- **After zig-zag scan, AC coefficients** $\rightarrow$ pairs:

$$(r, k)$$

- $r \rightarrow$ run of preceding zeros.
- $k \rightarrow$ category of non-zero amplitude.
- **Special symbols**:

$$(15, 0) = \text{ZRL, run of 16 zeros}$$

$$(0, 0) = \text{EOB, End Of Block}$$

- **EOB** $\rightarrow$ all remaining block coefficients are zero.

### Domanda 25

(TC&JPEG) Describe the JPEG lossless coding process applied to the following table of quantized DCT coefficients:

$$\begin{bmatrix} 10 & 3 & 0 & 0 & 0 & 0 & 0 & 0 \\ -2 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- Treat missing positions as zero-padded $8 \times 8$ block.
- **DC**:

$$DC_n = 10$$

- **JPEG sends DC dif and only iference**:

$$\Delta DC = DC_n - DC_{n-1}$$

- First block with $DC_{n-1} = 0 \rightarrow \Delta DC = 10$.
- **Non-zero AC in zig-zag order**:

$$3, \ -2, \ 1, \ 1, \ 4, \ 1$$

- **Run/category**:
  - $3 \rightarrow$ run 0, category 2
  - $-2 \rightarrow$ run 0, category 2
  - $1 \rightarrow$ run 0, category 1
  - $1 \rightarrow$ run 0, category 1
  - $4 \rightarrow$ run 6, category 3
  - $1 \rightarrow$ run 1, category 1
  - **EOB** $\rightarrow$ run 0, category 0
- **No ZRL** $\rightarrow$ no run longer than 15 zeros before non-zero coefficient.

**Domanda 26**

(TC&JPEG) What is the primary purpose of an orthogonal transform in the context of the transform coding paradigm?

- **Primary purpose** $\rightarrow$ **sparsification / energy compaction**.
- Orthogonal transform preserves energy and squared error:

$$T^{-1} = T^T, \quad \|TX\|^2 = \|X\|^2$$

**Domanda 27**

(TC&JPEG) In the context of JPEG compression, what is the role of the quantization table?

- **Quantization table** $\rightarrow$ quantization step for each DCT coefficient.
- Controls rate-quality tradeoff.
- **Reflects visual sensitivity**:
  - **low frequencies** $\rightarrow$ smaller steps;
  - **high frequencies** $\rightarrow$ larger steps.

**Domanda 28**

(TC&JPEG) Which statement regarding the relationship between the Arithmetic Mean (AM) and Geometric Mean (GM) of variances in transform coding is correct?

- Orthogonal transforms preserve energy → preserve arithmetic mean of coefficient variances.
- They do not necessarily preserve geometric mean.
- **Transform coding goal** → reduce GM while AM stays fixed, because optimal distortion depends on GM.

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# 4 Wavelet-Based Image Coding

**Domanda 29**

State the time-frequency uncertainty principle ($\Delta t \cdot \Delta f \geq 1/4\pi$), explain why it imposes a trade-off, and how wavelets address it with adaptive multiresolution (short windows at high frequencies, long at low frequencies).

$$\Delta t \cdot \Delta f \geq \frac{1}{4\pi}$$

- **Short window** $\rightarrow$ good time/space localization, poor frequency resolution.
- **Long window** $\rightarrow$ good frequency resolution, poor localization.
- **Wavelets** $\rightarrow$ multiresolution:
  - **high frequencies** $\rightarrow$ short basis functions;
  - **low frequencies** $\rightarrow$ long basis functions.
- **Image match** $\rightarrow$ edges need localization; smooth trends need coarse-scale representation.

**Domanda 30**

Compare STFT (rigid tiling) and DWT (adaptive tiling) of the time-frequency plane, linking them to the trends vs anomalies image model.

- **STFT** $\rightarrow$ fixed window size $\rightarrow$ rigid time-frequency tiling; same resolution for all frequencies.
- **DWT** $\rightarrow$ filter banks + downsampling $\rightarrow$ low-frequency approximations + high-frequency detail subbands.
- **Repeated DWT** $\rightarrow$ multiresolution representation.
- **Image model**:
  - **smooth trends** $\rightarrow$ coarse low-frequency components;
  - **anomalies/edges** $\rightarrow$ high-frequency localized details.

**Domanda 31**

(TC&JPEG) What is a primary advantage of the hierarchical (multiresolution) decomposition offered by the Wavelet Transform in JPEG2000?

- **Wavelet decomposition** $\rightarrow$ progressive and multiresolution coding.
- Low-resolution approximation decoded first $\rightarrow$ detail subbands refine quality and resolution.

## Domanda 32

(TC&JPEG) Compare the block-based DCT approach used in JPEG with the wavelet-based decomposition used in JPEG2000, specifically in terms of how they handle the image signal and the resulting artifacts.
(TC&JPEG) Which of the following is a key reason why JPEG2000 is generally more efficient than JPEG?

- **JPEG**:
  - independent $8 \times 8$ block DCT;
  - simple and efficient;
  - **low bitrate** $\rightarrow$ **blocking artifacts**.
- **JPEG2000**:
  - DWT over large tiles / whole image;
  - multiresolution representation;
  - embedded bitplane coding and precise rate control;
  - **low bitrate** $\rightarrow$ ringing/blurring near edges, not blocking.
- **Efficiency reasons**:
  - wavelet multiresolution decomposition;
  - embedded bitstream truncation;
  - independent codeblocks;
  - quality/resolution scalability;
  - no fixed $8 \times 8$ boundaries.

## Domanda 33

Describe the JPEG2000 architecture (Tier 1: DWT 9/7 or 5/3 -> fine quantization -> arithmetic coding of codeblocks per bitplane; Tier 2: EBCOT). Where does the lossy operation actually happen?

- **JPEG2000** $\rightarrow$ wavelet still-image codec for scalability, rate control, ROI, lossy/lossless.
- **Tier 1**:

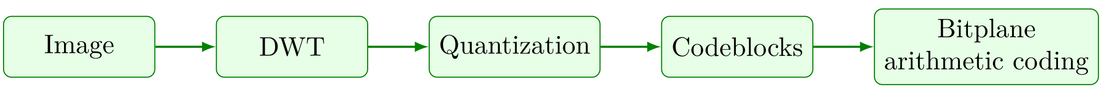


Figure 6: JPEG2000 Coding Pipeline

- **9/7 DWT** $\rightarrow$ irreversible, lossy.
- **5/3 DWT** $\rightarrow$ reversible, lossless.

- **Coefficients** → codeblocks → coded bitplane by bitplane.
- **Tier 2 / EBCOT** → layers, packets, progression orders, truncation points.
- **Lossy operations** → quantization and embedded bitplane/pass truncation.

**Domanda 34**

Compare JPEG and JPEG2000 on channel-error robustness (codeblock independence, contained vs catastrophic propagation, resynchronization markers).

- **JPEG**:
  - sequential entropy coding;
  - bit error can desynchronize Huffman stream;
  - corruption propagates until restart/resynchronization marker.
- **JPEG2000**:
  - relatively independent codeblocks;
  - stream organized into packets/layers;
  - damage tends to remain local to codeblock/subband/quality layer.
- **Result** → JPEG2000 more robust to localized channel errors.

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# 5 Learned Image Coding

**Domanda 35**

(TC&JPEG) Explain the fundamental shift in Rate-Distortion (R-D) optimization from classical codecs (e.g., JPEG) to neural compression methods.

- **Classical codecs** → optimize handcrafted tools:
  - fixed transforms;
  - prediction modes;
  - block partitions;
  - quantization steps.

$$J = D + \lambda R$$

- **Neural compression** → learns analysis transform, synthesis transform, entropy model.

$$\mathcal{L} = D(x, \hat{x}) + \lambda R(\hat{y})$$

- **Main shift** → data-learned transform instead of fixed standard tools.

**Domanda 36**

**JPEG-AI**: goals (beat VVC-Intra by about 50%), backbone (hierarchical VAE with hyperprior), dual-use human/machine support, and complexity profiles (Dec0/Dec1/Dec2, kMAC/px). Cite pros/cons vs classical codecs (R-D vs computational cost, determinism, hallucinations).

- **JPEG-AI** → learned still-image coding for human viewing and machine consumption.
- **Goal** → beat classical intra codecs, target about **50% gain vs VVC-Intra** in favorable settings; notes also report about **27% saving at 0.5 bpp**.
- **Backbone**:

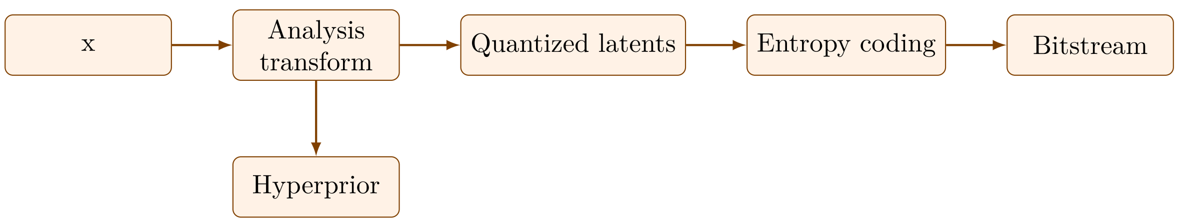


Figure 7: JPEG-AI Auto-Encoder Framework

- **Hyperprior** → side information about latent statistics (scales/probabilities); costs bits but improves entropy modeling.
- **Backbone class** → hierarchical **VAE** with hyperprior; nonlinear analysis/synthesis transforms learned end-to-end.

- **Dual use** → support both **human viewing** and **machine consumption**.
- Decoder profiles, complexity in **kMAC/px**:
  - **Dec0** → about 8 kMAC/px;
  - **Dec1** → about 23 kMAC/px;
  - **Dec2** → about 214 kMAC/px.
- **Pros**:
  - better R-D efficiency;
  - nonlinear transforms learned from data;
  - possible human/machine task support.
- **Cons**:
  - high computational/energy cost;
  - harder deterministic reproducibility;
  - risk of plausible but inaccurate reconstructed details.

**Domanda 37**

(TC&JPEG) What is the primary purpose of adding Additive Uniform Noise during the training phase of a neural codec?

- **Quantization** → non-dif and only iferentiable, zero gradient almost everywhere.
- **Training relaxation** → additive uniform noise:

$$u \sim \mathcal{U}\left(-\frac{1}{2}, \frac{1}{2}\right)$$

- **Effect** → approximates rounding with continuous operation → gradient-based optimization possible.

**Domanda 38**

(TC&JPEG) Why do Convolutional Neural Networks (CNNs) outperform Multi-Layer Perceptrons (MLPs) when applied to image compression?

- **MLPs** → flatten images, ignore spatial locality, many parameters, weak image bias.
- **CNNs** → exploit local correlations, translation structure, shared filters.
- **Result** → more efficient image compression models.

—

# 6 Audio and Speech Coding

**Domanda 39**

Explain masking in the auditory system: define frequency masking and temporal masking (pre/post-masking) with their time-scale orders of magnitude.

- **Auditory masking** → strong sound makes weaker sound inaudible.
- **Codec use** → hide quantization noise below masking threshold.
- **Frequency masking** → simultaneous masking; strong component at $f_0$ raises threshold around nearby frequencies, mostly same critical band.
- **Temporal masking**:
  - **pre-masking** → few ms before masker;
  - **post-masking** → up to about 100-200 ms after masker, depending on level/content.

**Domanda 40**

What is the critical band and how does it relate to the audibility condition of a set of sinusoids close in frequency?

- **Critical band** → frequency interval processed roughly by same cochlear auditory filter.
- Bandwidth increases with center frequency.
- Close sinusoids interact inside same band.
- **Audibility condition**:

$$\sum_i g_i^2 > S_0(f_n)$$

- $g_i^2$ → sinusoid powers.
- $S_0(f_n)$ → absolute hearing threshold near frequency $f_n$.

**Domanda 41**

(A&S Comp.) Explain the dif and only iference between Source-Based (Parametric) coding and Sink-Based (Perceptual) coding.

- **Source-based / parametric coding** → models signal production.
  - **Speech** → source-filter model of vocal folds + vocal tract.
  - **Goal** → intelligibility and low latency at very low bitrate.
- **Sink-based / perceptual coding** → models listener perception.

- **Audio codecs** → auditory masking + critical bands.
- **Goal** → transparent quality by removing inaudible information.

**Domanda 42**

(A&S Comp.) Describe the "Analysis-by-Synthesis" (AbS) loop used in CELP (Code Excited Linear Prediction) codecs and why it represents an improvement over simple LPC-10.

- **LPC-10** → simplified excitation: impulse train for voiced, noise for unvoiced → synthetic sound.
- **CELP** → **Analysis-by-Synthesis** closed-loop search:

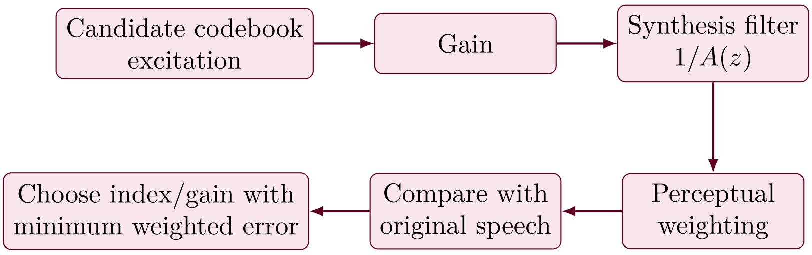


Figure 8: CELP Analysis-by-Synthesis Loop

- **Bit allocation**:
  - **sensitive bands** → more bits;
  - **masked bands** → fewer bits.

**Domanda 44**

(A&S Comp.) Describe the principles of the LPC10 speech coding scheme.

- **LPC** → short speech frame modeled as excitation through all-pole vocal-tract filter.
- **Encoder sends** → filter parameters, gain, excitation information.

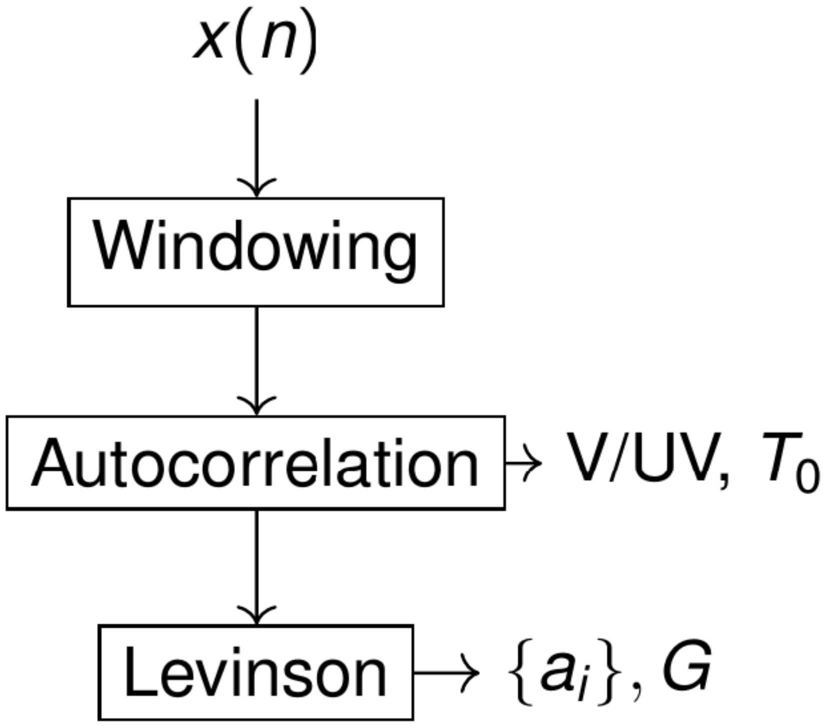


Figure 9: Linear predictive coding - analysis.png

- **Analysis**:

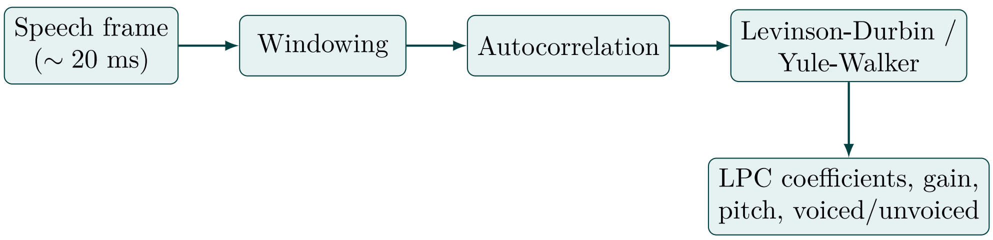


Figure 10: LPC10 Speech Analysis Chain

- **Prediction model**:

$$\hat{x}(n) = -\sum_{i=1}^{P} a_i x(n-i)$$

- **Residual**:

$$y(n) = x(n) - \hat{x}(n) = \sum_{i=0}^{P} a_i x(n-i), \quad a_0 = 1$$

- **Voiced frames** → pitch-period excitation.
- **Unvoiced frames** → noise-like excitation.
- **LPC-10** → low bitrate, less natural than CELP because excitation is rigid.

**Domanda 45**

(A&S Comp.) Draw the scheme and describe the operation of the functional blocks of an MP3 encoder.

- **MP3** → perceptual audio codec.

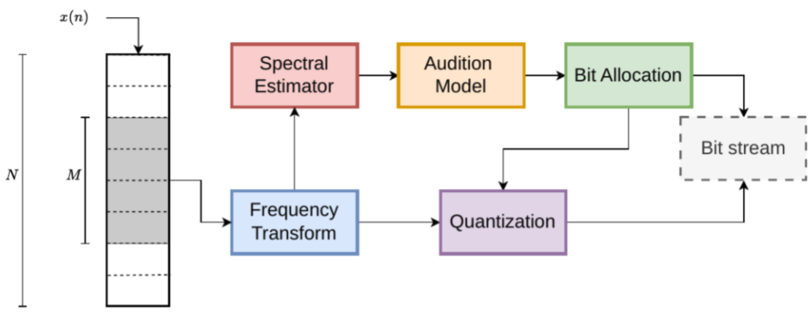


Figure 11: Pasted image 20260624185047.png

- **Main encoder chain**:

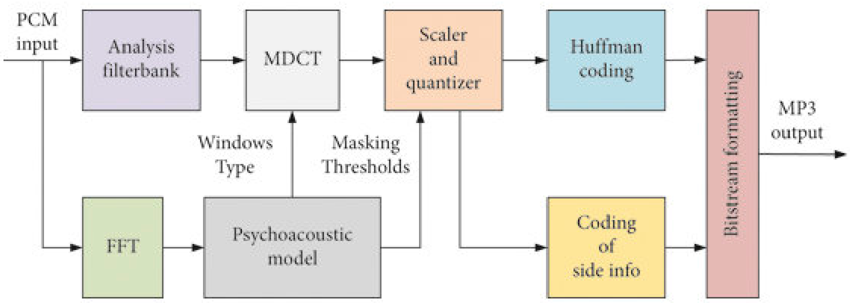


Figure 12: Pasted image 20260624213214.png

- **Functional blocks**:
  - framing/windowing + transform/filterbank → convert time signal into spectral components;
  - **psychoacoustic model** → estimate masking threshold from signal spectrum;
  - **bit allocation** → spend more bits where ear is sensitive, fewer where noise is masked;
  - **quantization** → lossy stage on spectral coefficients;
  - **entropy coding** → final lossless compression stage.

- **Psychoacoustic model** $\rightarrow$ encoder-side only; not needed at decoder.
- **Decoder** $\rightarrow$ entropy decoding, inverse quantization, inverse transform / synthesis filterbank.

**Domanda 46**

(A&S Comp.) Why are Line Spectrum Frequencies (LSF) preferred over direct quantization of LPC coefficients ($a_i$)?

- **Direct LPC coefficient quantization** $\rightarrow$ can create unstable synthesis filters.
- **LSF** $\rightarrow$ represent LPC filter through roots on unit circle.
- Stability condition easier to enforce by preserving interlacing/order:

$$0 < \omega_1^{(P)} < \omega_1^{(Q)} < \omega_2^{(P)} < \cdots < \pi$$

- **If quantization disturbs order** $\rightarrow$ decoder can restore valid ordering.

**Domanda 47**

(A&S Comp.) Regarding the Opus audio codec, what is the primary technical advantage of its hybrid design?

- **Opus** combines:
  - **SILK** $\rightarrow$ LPC-based, efficient for speech and low bitrates;
  - **CELT** $\rightarrow$ MDCT-based, efficient for music/general audio and low delay.
- **Advantage** $\rightarrow$ seamless adaptation across speech/music, bitrate, latency; useful for WebRTC.

**Domanda 48**

(A&S Comp.) Which of the following best describes current research trends in the future of multimedia audio coding?

- Neural speech/audio codecs at very low bitrates.
- Perceptual and QoE-driven quality metrics.
- Spatial/immersive audio.
- Robust real-time coding under latency constraints.
- Quality assessment for generated/enhanced audio.

—

# 7 Quality Evaluation and Quantization

**Domanda 49**

**Compare MSE/PSNR, SSIM and LPIPS**: what each measures, pros/cons, and why two images with the same MSE can have very dif and only iferent perceived quality.

- **MSE** → average squared pixel error:

$$\mathrm{MSE} = \frac{1}{NM}\sum_{i,j}(x_{ij} - \hat{x}_{ij})^2$$

- **PSNR** → logarithmic MSE:

$$\mathrm{PSNR} = 10\log_{10}\left(\frac{MAX^2}{\mathrm{MSE}}\right)$$

- **MSE/PSNR** → simple and reproducible, but weak perceptually; ignore structure, masking, semantics.
- **SSIM** → compares local luminance, contrast, structure.
- **LPIPS** → compares deep features; better for perceptual/semantic distortions, but expensive and model-dependent.
- **Same MSE can look dif and only iferent** → error distribution matters: texture noise less visible; edge/structure error more annoying.

**Domanda 50**

(MMQuEv) Explain the dif and only iference between subjective and objective quality evaluation in multimedia systems and why both are necessary.

- **Subjective evaluation** → human observers rate perceived quality.
  - Ground truth for perception.
  - Slow, expensive, statistically variable.
- **Objective evaluation** → automatic signal/feature metric.
  - Fast, repeatable, scalable.
  - May correlate imperfectly with human perception.
- **Need both** → subjective validates perception; objective enables optimization and monitoring.

**Domanda 51**

(MMQuEv) What are the key stages involved in designing a subjective quality test according to standardized guidelines?

- **Define**:
  - dataset/content selection;
  - display/audio equipment;
  - viewing/listening conditions;
  - **methodology**: ACR, DSIS, pairwise comparison, etc.;
  - participant screening;
  - score processing/outlier handling.
- **Control factors** → viewing distance, brightness/contrast, room illumination, audio setup, test duration.

**Domanda 52**

(MMQuEv) Describe the main categories of objective quality metrics based on the availability of the original "reference" signal.
(MMQuEv) Which of the following best describes the "Full-Reference" (FR) objective quality assessment approach?

- **Full Reference (FR)** → original and degraded signals available; examples: MSE, PSNR, SSIM, VMAF.
- **Reduced Reference (RR)** → only compact original features available.
- **No Reference (NR)** → only degraded signal available; examples: BRISQUE, NIQE, PIQE.
- **FR luminance PSNR**:

$$\mathrm{MSE}_Y(t) = \frac{1}{NM}\sum_{n,m}[I(n,m,1,t) - \hat{I}(n,m,1,t)]^2$$

$$\mathrm{PSNR}_Y(t) = 10\log_{10}\left(\frac{255^2}{\mathrm{MSE}_Y(t)}\right)$$

**Domanda 53**

(MMQuEv) In the context of subjective testing, what is the main purpose of a "screening" phase for participants?

- **Screening** → checks participants can correctly perceive stimuli.
- **Removes invalid observers**: uncorrected visual issues, color-vision problems, inconsistent scoring behavior.

**Domanda 54**

(MMQuEv) Why is statistical analysis a critical component of subjective quality evaluation?

- **Human scores vary** → need reliability measures.
- **Statistical analysis** → MOS, variance, confidence intervals, outlier detection.
- **MOS**:

$$\text{MOS} = \frac{1}{N}\sum_{i=1}^{N} x_i$$

- **Standard error**:

$$SE = \frac{s}{\sqrt{N}}$$

**Domanda 55**

(S&P Qnt) Explain the dif and only iference between a "mid-tread" and a "mid-rise" quantizer in the context of uniform quantization for signed data.

- **Mid-tread** → zero is reconstruction level; small values around zero map to zero.
- **Mid-rise** → zero is decision threshold; values around zero map to positive/negative non-zero levels.
- **Residuals prefer mid-tread** → many small values become exactly zero:

$$Q(x) = \Delta \cdot \text{round}\left(\frac{x}{\Delta}\right)$$

**Domanda 56**

(S&P Qnt) Define the concept of a "deadzone" in a quantizer and explain why it is frequently employed in lossy compression systems.

- **Deadzone quantizer** → enlarged interval mapped to zero:

$$Q(x) = 0 \quad \text{for small } |x|$$

- **Useful in lossy compression** → prediction/transform residuals often have many small coefficients.
- **Consequence** → more zeros → better entropy coding.

**Domanda 57**

(S&P Qnt) Why is scalar quantization alone often considered insufficient for effective compression of non-sparse data?

- **Scalar quantization** → one sample at a time.

- **Non-sparse signal** → many significant samples remain → entropy coding has little to exploit.
- **Better compression needs**:
  - **prediction** → reduce residual variance;
  - **transform coding** → compact energy;
  - **vector/block coding** → exploit dependencies;
  - **entropy coding** → exploit non-uniform probabilities.

**Domanda 58**

(S&P Qnt) What is the condition for a predictive quantization system to be effective, and how is the "coding gain" defined?

- **Effective condition**:

$$\sigma_y^2 < \sigma_x^2$$

**with**:

$$y(n) = x(n) - v(n)$$

- **Prediction gain**:

$$G_P = 10\log_{10}\left(\frac{\sigma_x^2}{\sigma_y^2}\right)$$

- Positive when predictor reduces variance.

**Domanda 59**

(S&P Qnt) Draw the scheme of a linear predictive quantization scheme, and motivate the structure, with particular attention to the use of a decoding loop at the encoder side.

- Predictor must use reconstructed past samples on encoder and decoder:

$$v(n) = \mathcal{P}(\hat{x}(n-1), \hat{x}(n-2), \dots)$$

$$y(n) = x(n) - v(n)$$

$$\hat{x}(n) = \hat{y}(n) + v(n)$$

- Encoder includes same reconstruction loop as decoder → prevents drift.

### Domanda 60

(S&P Qnt) What is the primary purpose of the predictor in a predictive quantization system?
(S&P Qnt) In a predictive quantization system, if the prediction $v(n)$ is nearly equal to $x(n)$, what happens to the variance of the signal $y(n)$ being sent to the quantizer?

- **Predictor** $\rightarrow$ exploits correlation among neighboring samples.
- If $v(n) \approx x(n)$:

$$y(n) = x(n) - v(n) \approx 0$$

$$\sigma_y^2 \ll \sigma_x^2$$

- Quantizer encodes lower-energy residual.

### Domanda 61

(S&P Qnt) When selecting a linear predictor of order $P$ for a random process, how does the prediction error variance typically behave as the order increases?

- **Linear predictor**:

$$v(n) = -\sum_{i=1}^{P} a_i x(n-i)$$

$$y(n) = \sum_{i=0}^{P} a_i x(n-i), \quad a_0 = 1$$

- Increasing $P$ cannot worsen optimal error variance in theory $\rightarrow$ old solution still available.
- **Practice** $\rightarrow$ larger $P$ increases complexity, side information, estimation error; gains eventually negligible.

### Domanda 62

**Explain the screening effect**: why does the prediction gain saturate as the linear predictor order $P$ increases?

- Increasing $P$ $\rightarrow$ optimal prediction error variance cannot increase in theory.
- Gain usually saturates due to **screening effect**.
- Nearby samples already explain most correlation with current sample.
- Farther samples add little independent information because correlation is mostly mediated by nearer samples.

- **Result** → diminishing returns plus higher complexity/estimation sensitivity.

**Domanda 63**

(S&P Qnt) In high-resolution uniform quantization, what is the approximate relationship between the SNR and the bit rate ($R$)?

- High-resolution uniform quantization:

$$D \approx \frac{A^2}{12} 2^{-2R}$$

$$\text{SNR} = 10 \log_{10}\left(\frac{\sigma_X^2}{D}\right) = 10 \log_{10}\left(\frac{\sigma_X^2}{\frac{A^2}{12} 2^{-2R}}\right)$$

- With $\gamma^2 = \frac{A^2}{4\sigma_X^2}$:

$$\text{SNR} \approx 6.02R - 10 \log_{10}\left(\frac{\gamma^2}{3}\right)$$

- **Rule** → each extra bit/sample gives about 6 dB SNR improvement.

—

# 8 Adaptive Streaming

**Domanda 64**

**Compare QoS and QoE**: what they measure and how they correlate. Which network factors impact streaming QoE?

- **QoS** → technical service/network metrics: throughput, latency, jitter, packet loss.
- **QoE** → user-perceived service quality: media quality, startup delay, stalls, quality switches, device, context, expectations.
- **Correlation** → not linear.
  - Packet loss may be hidden by buffer.
  - Short throughput drop may cause rebuffering → large QoE loss.

**Domanda 65**

(AdapStrm) Describe the fundamental architectural dif and only iferences between a "Push-based" streaming system (e.g., RTP/UDP) and a "Pull-based" system (e.g., DASH).

- **Push-based streaming** (RTP/UDP/WebRTC-style):
  - server sends packets according to timing;
  - low latency;
  - needs real-time transport/control;
  - **typical use** → conferencing/live interactive media.
- **Pull-based streaming** (DASH/HLS over HTTP):
  - client requests segments;
  - works with HTTP/CDNs/caches;
  - client selects bitrate from throughput/buffer;
  - higher delay, scalable/robust for VoD/streaming.

**Domanda 66**

(AdapStrm) Analyze the role of the client-side buffer in the context of stability and Quality of Experience (QoE).

- **Buffer** → stores downloaded media in seconds.
- **Function** → absorbs throughput variation and jitter.
- **Larger buffer** → more stability, but higher startup delay/latency.

- QoE depends on startup time, stalls, stall duration, segment quality, quality switches.

**Domanda 67**

(AdapStrm) Define “Switching Penalty” and discuss its impact on perceived video quality.

- **Switching penalty** → QoE loss from visible quality changes between segments.

$$J(n) = \lambda_1 K_n - \lambda_2 |K_n - K_{n-1}| - \phi(\Delta_n)$$

- $K_n$ → segment quality.
- $|K_n - K_{n-1}|$ → switching penalty.
- $\phi(\Delta_n)$ → rebuffering penalty.
- Frequent oscillations can be worse than stable slightly lower quality.

**Domanda 68**

(AdapStrm) Explain the evolution of the playout buffer level $B(t)$ using a mathematical model. In your explanation, describe the dynamics of the “playback” (draining) phase and the “rebufferization” (stalling) phase.

- **Buffer level in playback seconds**:

$$B(t) = L \cdot T_s$$

- $L$ → stored segments; $T_s$ → segment duration.
- Coding rate $R_c$, throughput $S$:

$$\frac{dB}{dt} = \begin{cases} \frac{S}{R_c} - 1 & \text{during playback} \\ \frac{S}{R_c} & \text{during rebuffering} \end{cases}$$

- **Playback** → buffer receives data but drains at one playback second per real second.
- **Rebuffering** → playout stops; no output drain.
- **Stable playback average**:

$$S > R_c$$

**Domanda 69**

(AdapStrm) What is the primary motivation for using HTTP-based protocols for video streaming?

- **HTTP streaming** → stateless, CDN-friendly, firewall-friendly, easy to deploy at scale.

- **DASH/HLS** → use web infrastructure; client adapts representation quality.

**Domanda 70**

(AdapStrm) What is the consequence of an ABR algorithm that systematically overestimates the available bandwidth?

- **ABR overestimates throughput** → requests too high bitrate.
- **Segment download time grows** → buffer empties → rebuffering/stalling.

**Domanda 71**

(AdapStrm) Which metric is a direct indicator of streaming QoE from the end-user's perspective?

- **Direct QoE indicators** → startup delay, rebuffering duration/frequency, quality level, quality switches.
- **User perspective** → uninterrupted stable playback often more important than low-level network metrics.

**Domanda 72**

(AdapStrm) At what stage in the session lifecycle does a DASH client process the Media Presentation Description (MPD)?

- Client processes **MPD** at session start, before media segment download.
- MPD lists periods, adaptation sets, representations, codecs, bitrates, resolutions, segment URLs.

**Domanda 73**

Explain how an ABR (Adaptive Bitrate) algorithm works: rate-based vs buffer-based logic, and the risk of overestimating available bandwidth.

- **ABR** → client-side rule selecting bitrate representation for next segment.
- **Goal** → maximize QoE: quality + stability, avoid stalls.
- **Rate-based**:
  - estimate recent throughput;
  - choose representation below available bandwidth, often with safety margin.

- **Buffer-based**:
  - **low buffer** → conservative bitrate;
  - **high buffer** → higher quality.
- **Overestimated bandwidth** → segments too large → long downloads → buffer drain → stalls.

—

# 9 Motion Estimation and Video Coding

**Domanda 74**

**Motion estimation**: give the principles of the block matching approach. Give at least one cost function. [Bonus]. Discuss the regularization issue.

- **Block matching** → split frame into blocks $B_{p,q}$; search similar block in reference frame window.
- **Best displacement**:

$$(\hat{i}, \hat{j}) = \arg \min_{(i,j) \in W} J(i, j)$$

- **SSD**:

$$J_{\text{SSD}}(i, j) = \sum_{(n,m) \in B_{p,q}} [f(n, m, k) - f(n - i, m - j, h)]^2$$

- **SAD**:

$$J_{\text{SAD}}(i, j) = \sum_{(n,m) \in B_{p,q}} |f(n, m, k) - f(n - i, m - j, h)|$$

- **Regularized**:

$$J_{\text{REG}}(i, j) = \|\vec{f}_k(B_{p,q}) - \vec{f}_h(B_{p-i,q-j})\|_p^p + \lambda R(i, j)$$

- **Regularization** → penalizes expensive/irregular motion vectors; balances prediction quality with motion-vector coding rate.

**Domanda 75**

Discuss the advantages and the disadvantages of Intra, Predictive, and Bidirectional images in a GOP for video coding.

- **I images**:
  - intra only;
  - **advantages** → random access, error reset, no motion estimation;
  - **disadvantages** → high rate, lower compression.
- **P images**:
  - predicted from previous anchor frames;
  - **advantages** → good compression, moderate delay;
  - **disadvantages** → ME/MC complexity, error propagation.
- **B images**:
  - predicted from past and future frames;

  - **advantages** $\rightarrow$ best compression;
  - **disadvantages** $\rightarrow$ highest complexity, structural delay, not ideal for low latency.

- GOP structure controls compression efficiency, delay, random access interval, error propagation.

**Domanda 76**

(ME) Describe the dif and only iference between "motion field" and "optical flow".

- **Motion field** $\rightarrow$ projection of actual 3D scene motion onto image plane.
- **Optical flow** $\rightarrow$ apparent motion of brightness patterns.
- They often coincide, but can dif and only ifer due to illumination changes, occlusions, texture ambiguities.

**Domanda 77**

(ME) Explain the Horn and Schunck algorithm's core principle for dense optical flow estimation.

- **Optical flow constraint**:

$$uf_x + vf_y + f_t = 0$$

- **Horn-Schunck objective**:

$$J = \iint_{\mathcal{R}} (uf_x + vf_y + f_t)^2 \, dxdy + \lambda \iint_{\mathcal{R}} (\|\nabla u\|^2 + \|\nabla v\|^2) \, dxdy$$

- **First term** $\rightarrow$ brightness consistency.
- **Second term** $\rightarrow$ smooth dense flow field.

**Domanda 78**

(ME) Discuss the Rate-Distortion trade-off when selecting block sizes in motion estimation.

$$J(v) = d(B_k^{(p)}, B_h^{(p+v)}) + \lambda_{ME} R(v)$$

- **Large blocks**:
  - fewer motion vectors;
  - lower side information and complexity;
  - worse fit near object boundaries.
- **Small blocks**:

- better prediction and lower distortion;
- more motion vectors and partition bits;
- higher complexity.

**Domanda 79**

(ME) Which of the following is a disadvantage of using the Sum of Squared Dif and only iferences (SSD) as a matching criterion?

- **SSD squares errors** → outliers dominate cost.
- Illumination changes, noise, occlusions → irregular motion vectors.
- **SAD** → often more robust and cheaper.

**Domanda 80**

(ME) What is the main benefit of the Hexagon Search strategy compared to Full Search?

- **Full search** → tests all candidates; optimal within window, expensive.
- **Hexagon search** → tests small hexagonal pattern iteratively; far fewer evaluations.
- **Benefit** → faster search.
- **Cost** → no guaranteed global optimum.

**Domanda 81**

(ME) What does an affine motion model allow that a pure translational model does not?

- **Translation** → constant displacement only.
- **Affine motion** → rotation, zoom, shear:

$$\vec{v}(p) = \vec{b} + Bp = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} + \begin{bmatrix} b_3 & b_4 \\ b_5 & b_6 \end{bmatrix} p$$

- If $B = 0$ → pure translation / block matching.

**Domanda 82**

Draw the block diagram of a hybrid video encoder (motion estimation/compensation + DCT + quantization + entropy coding + reconstruction loop with frame buffer). Explain why the encoder contains an internal decoder.

- **Hybrid video encoder** → codes prediction residual, usually not whole frame.

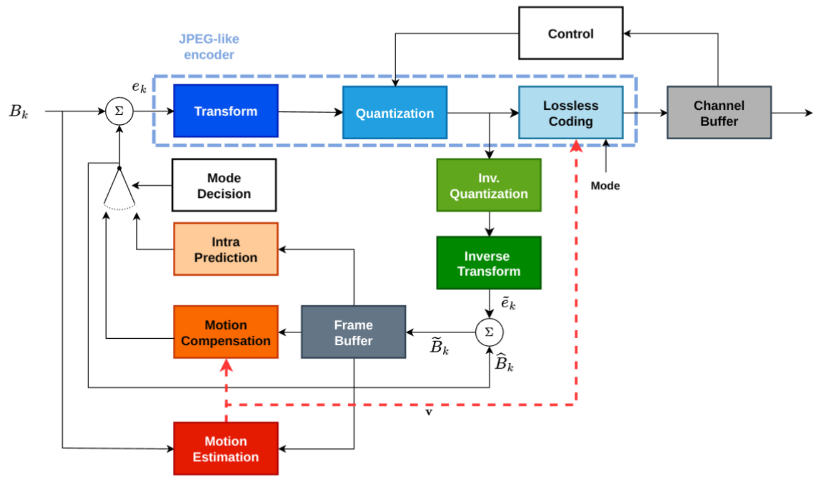


Figure 13: Hybrid video encoder.png

- **Mode decision** → selects intra/inter predictor.
- **Transform + quantization** → residual coding.
- **Lossless coding** → bitstream.
- Inverse quantization + inverse transform → reconstructed residual.
- **Reconstructed block** → frame buffer for later motion compensation.
- **Internal decoder needed** → future predictions must use same reconstructed frames as decoder; otherwise drift propagates.

**Domanda 83**

(VCP) Why is the temporal prediction error usually more efficient to encode than the original video signal?

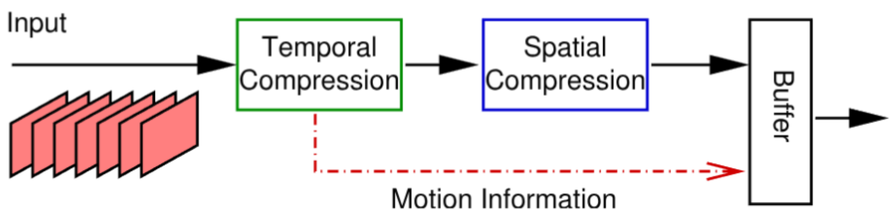


Figure 14: Video coding principles.png

- **Video frames** → highly correlated.
- Motion-compensated prediction estimates current block from reference frames.

- **Encode only residual**:

$$e = x - x_p$$

- **Residual** → lower energy and more sparse than original frame.
- **Consequence** → transform quantization and entropy coding more efficient.

**Domanda 84**

(VCP) Describe the function of the “Mode Selection” step in a hybrid video encoder.

- **Mode selection** → choose coding mode minimizing R-D cost:

$$J = D + \lambda R$$

- **For blocks**:

$$D = \sum_{k=1}^{K} D_k(i_k, Q), \quad R = \sum_{k=1}^{K} R_k(i_k, Q)$$

$$J(\vec{i}, Q, \lambda) = D(\vec{i}, Q) + \lambda R(\vec{i}, Q)$$

- **Block-wise suboptimal minimization**:

$$i_k^\star = \arg\min_{i_k} J_k(i_k, Q, \lambda)$$

- **Empirical choices**:

$$\text{MPEG-2: } \lambda = aQ^2 + b$$

$$\text{H.264: } \lambda = c \cdot 2^{dQ+e}, \quad \lambda_{ME} = \sqrt{\lambda}$$

**Domanda 85**

(VCP) How does the “Channel Buffer” controller manage the trade-off between target rate and video quality?

- **Channel buffer** → controls quantization step to meet target rate.
- **Buffer occupancy high** → increase quantization step → reduce rate, lower quality.
- **Buffer occupancy low** → decrease quantization step → improve quality, higher rate.
- **Goal** → avoid overflow/underflow while keeping quality high.

**Domanda 86**

(VCP) What is the primary role of an “I-frame” in a GOP structure?

- **I-frame** → intra-coded and independently decodable.
- **Roles** → random access, GOP start, temporal error-propagation limit.

**Domanda 87**

(VCP) In the context of motion vector coding, why is a Median Predictor (MVP) used?

- **Neighboring motion vectors** → correlated.
- Median predictor estimates current vector from adjacent blocks.
- **Encoder sends only dif and only iference**:

$$\mathrm{MVD} = \mathrm{MV} - \mathrm{MVP}$$

- **MVD usually small** → cheaper to encode.

**Domanda 88**

(VCP) What happens in the decoder when it receives an "Inter-coded" block?

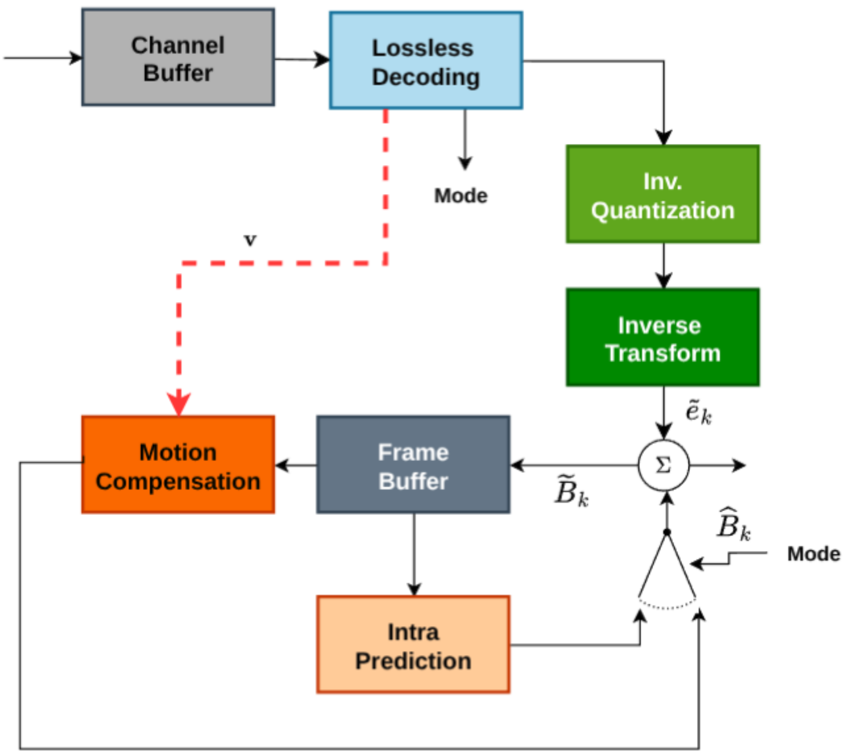


Figure 15: Hybrid video decoder.png

- **Decoder reads side information** → mode, reference index, motion vector.

- **Entropy decoding** $\rightarrow$ residual and side information.
- Inverse quantization + inverse transform $\rightarrow$ reconstructed residual.
- Motion compensation / intra prediction $\rightarrow$ predictor.
- **Reconstruction**:

$$\hat{x} = \text{prediction} + \hat{e}$$

- Output stored in frame buffer.

—

# 10 Modern Video Coding

**Domanda 89**

(ModernVC) What is the specific scope of video compression standards like H.266/VVC?

- **Standards define**:
  - bitstream syntax;
  - normative decoder behavior.
- They do not prescribe full encoder implementation.
- **VVC** → higher compression efficiency than HEVC, especially high-resolution/HDR/360/immersive video; cost → much higher complexity.

**Domanda 90**

(ModernVC) Explain the advantage of the "Coding Tree Unit" (CTU) structure introduced in HEVC/VVC.

- **CTU** → replaces fixed macroblocks with flexible recursive partitioning.
- **Large homogeneous regions** → large blocks.
- **Detailed regions** → smaller blocks.
- **Result** → improved R-D efficiency.
- **HEVC** → quadtree partitioning.
- **VVC** → multi-type trees, including binary and ternary splits.

**Domanda 91**

(ModernVC) What are the roles of VCL and NAL in modern video standards?

- **VCL (Video Coding Layer)** → compressed video data: slices, prediction, residuals, coding syntax.
- **NAL (Network Abstraction Layer)** → wraps VCL/non-VCL data into NAL units for transport/storage.
- **Purpose** → separates codec syntax from packetization.
- **Non-VCL examples** → SPS, PPS, SEI.

**Domanda 92**

(ModernVC) What is the main purpose of the CABAC entropy coder in modern standards?

- **CABAC** = Context-Adaptive Binary Arithmetic Coding.
- **Workflow** → binarize syntax elements, adapt probabilities to local context, arithmetic-code bins.
- **Purpose** → lossless entropy coding with better compression than VLC/CAVLC.
- **Cost** → higher complexity.

**Domanda 93**

(ModernVC) Why are “Tiles” considered “hardware-friendly” in VVC and HEVC?

- **Tiles** → independently decodable rectangular frame regions.
- Restrict dependencies across tile boundaries.
- Enable parallel encoding/decoding on CPU/GPU cores.
- **Result** → lower synchronization cost.

**Domanda 94**

(ModernVC) What is the function of an “In-Loop Filter” like the Adaptive Loop Filter (ALF)?

- **In-loop filters** → applied inside reconstruction loop before frames are stored as references.
- **Purpose** → reduce artifacts before motion compensation reuses frames.
- **Evolution**:
  - **H.264** → deblocking;
  - **HEVC** → SAO;
  - **VVC** → ALF and other tools.
- **ALF** → adaptive filtering of reconstructed samples; reduces ringing/residual distortion.

**Domanda 95**

Describe the intra-coding modes in H.264. [Optional] Discuss also the Intra modes in H.265.

- **Intra prediction** → exploits already reconstructed neighboring samples in same frame.
- **H.264/AVC**:
  - $4 \times 4$ blocks → 9 modes: 8 directional + DC;
  - $16 \times 16$ luma blocks → 4 modes.

- **H.265/HEVC** → 35 modes: 33 directional + DC + Planar.
- **H.266/VVC** → 65 directional modes + wide-angle variants for non-square blocks.
- **MPM lists** → predict likely mode from neighboring blocks → reduce signaling cost.

**Domanda 96**

Describe the principle of the deblocking in-loop filter of H.264.

- Low bitrate block transform + quantization → visible block-boundary discontinuities.
- **H.264** → normative in-loop deblocking filter on edges between $4 \times 4$ blocks.
- **Filtering strength depends on**:
  - coding mode;
  - motion vectors;
  - reference frames;
  - quantization parameter;
  - local boundary conditions.
- **In-loop reason** → filtered reconstructed frames become future references → avoids propagating blocking artifacts.